# Nate Cirino

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Available May 2027

## Education

**Northeastern University** May 2027 | MGPA: 3.6/4.0

*B.S. in Computer Science, Systems Concentration* *Boston, MA*

**Coursework:** Computer Systems, Digital Design & Computer Architecture, Computer Networks, Algorithms & Data Structures, Object-Oriented Design, Foundations of Software Engineering, Systems Security, Foundations of Data Science, Cybersecurity

## Experience

**Wayfair** Jul 2026 – Present

*Software Engineering Co-op* *Boston, MA*

- Own front-of-house device stack across four Wayfair retail brands: e-labels, mobile, and in-store POS
- Built supplier-agnostic e-label API unifying 4 vendor protocols behind one interface for outlet launches
- Shipped POS checkout flows with live device-to-backend customer sync and in-store scan-to-basket switching

**Wolters Kluwer** Jul 2025 – Jun 2026

*Software Engineer* *Boston, MA*

- Overhauled Relay JMS broker serialization with json-io across 20+ microservices, eliminating timeouts at scale
- Optimized Expert AI search and filtering across 20+ microservices, improving clinical query relevance at scale
- Built Java Spring microservices for CME credit issuance in Expert AI, serving 250K+ clinicians
- Implemented OIDC/IdP authentication via Azure across 20 enterprise health systems, securing 10K+ clinicians

**SculptAI** Jan 2025 – Aug 2026

*Software Architect* *New York, NY*

- Maintained 99.99% uptime across 1-2M monthly model calls for 2K paying users via AWS WAF, Datadog, and K8s
- Architected Flask backend with RAG pipeline for personalized workout recommendations using Bedrock KB
- Led eng org restructure at $1M-funded startup, partitioning teams into DevOps, dev, and agentic workflows

**Chevron New Energies** May 2024 – Aug 2024

*Software Engineer Intern* *Houston, TX*

- Implemented CI/CD workflows in Azure DevOps and K8s, automating deployment and rollback across services
- Built RESTful APIs and data pipelines using Azure Data Factory, improving product efficiency by 40%

## Projects

**NOAA AUV Sonar Detection** | *Python, PyTorch, AWS Neuron SDK, SageMaker* April 2025 – July 2025

- Trained CNN models on thermal sensor spectrograms to detect novel aquatic signatures for NOAA autonomous AUVs
- Designed AWS Neuron SDK deployment path for onboard custom-silicon inference under hard real-time limits
- Hardened architecture and loss formulation against sensor noise and class imbalance, improving detection

**Kepler** | *Rust, C++, Raft, LSM-tree, MVCC, Tokio* Mar. 2026 – Present

- Built a distributed, linearizable key-value store in Rust with from-scratch Raft consensus and LSM-tree engine
- Implemented Raft election, log replication, and membership changes; storage: memtable, WAL, leveled SSTables
- Added MVCC layer giving snapshot isolation and linearizable reads across a concurrent multi-node Raft cluster
- Built single-tier compaction with full SSTable rewrite every 4 flushes; inline, writer-blocking execution

**Sentinel** | *Next.js, FastAPI, Claude, pgvector, OpenTelemetry, Docker* Apr. 2026 – Present

- Built an AI SRE agent autonomously triaging incidents via tool-use over Prometheus, Tempo, and Loki telemetry
- Human-in-the-loop remediation gated behind approval; running live in SculptAI production triaging incidents

## Skills & Honors

**Languages/Tools:** C++, Rust, C, Python, Linux, Bash, Java, TypeScript, JavaScript, Docker, Kubernetes, AWS, Azure, Terraform, Jenkins, Buildkite, Groovy, PostgreSQL, GraphQL

**Frameworks & Libraries:** PyTorch, TensorFlow, Tokio, FastAPI, Flask, Pandas, Node.js, Express, React, Next.js

**AI / Agentic Tooling:** Devin, Cursor, Claude Code, Claude Cowork, GitHub Copilot, CodeRabbit

**Honors:** Tech Lead, Northeastern Electronic Racing; Dean's List (All Semesters); json-io OSS Contributor; PawHacks